MILITARY SPECIFICATION

RACK, TEST TUBE, LABORATORY, EPOXY-COATED WIRE, 40 TUBE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of epoxy-coated steel wire rack, suitable for holding 40 test tubes of 3/4 inch diameter (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Set-Up.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 S. 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6640

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3. REQUIREMENTS

3.1 Material. The rack shall be constructed of carbon steel wire with an epoxy coating.

3.2 Construction. Prior to epoxy coating construction shall be in accordance with Figure 1. The shelves shall be so aligned that the tubes will stand in vertical position. All wire intersections shall be spotwelded. Wire ends shall be as shown on Figure 1 or shall be tightly and neatly crimped around the frame. Wires of bottom frame grid shall be adequately spaced to properly support each tube in a vertical position. Sharp edges and burrs shall be removed. All openings shall be uniform and equally spaced except end row openings which may differ due to frame corner radius.

3.3 Spotwelding. The spotwelds shall be free from cracks or porosity.

3.4 Coating. The entire rack shall be epoxy coated. Coating shall have no cracks or porosity. There shall be no signs of steel dissolving when tested as specified in 4.4.1.

3.5 Dimensions. All dimensions prior to coating shall be in accordance with Figure 1. Thickness of steel wire referenced in this specification are nominal standard sizes. Commercial tolerances shall apply. After coating all hole dimensions shall allow 3/4 inch test tubes to be inserted with clearance all around the tube.

3.6 Workmanship. Workmanship shall be first class throughout. Racks shall be free from defects which detract from their appearance or may impair their serviceability.

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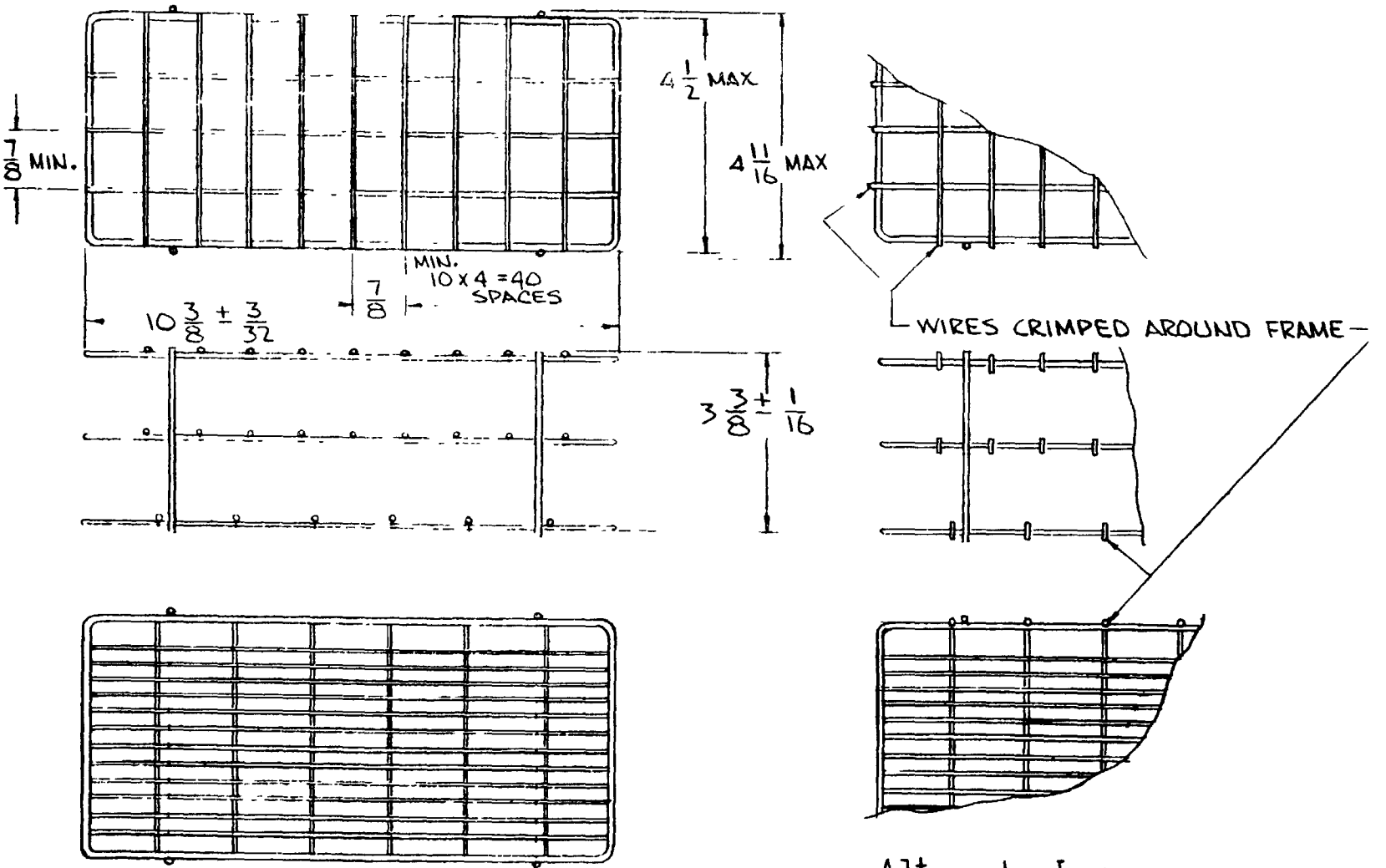


FIGURE 1 - Rack, test tube, laboratory

NOTE: Material
Carbon steel wire
[illegible] and supports .135 dia.
Grids .058 [illegible].

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, or auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table 1. The unit of product shall be one rack.

TABLE I. For sampling.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-3	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II.

TABLE II. For tests.

Characteristics	Requirement	Test procedure	Inspection level	AQL
Continuity of coating	3.4	4.4.1	S-3	*

* Acceptance number shall be zero.

4.3 Examination. The racks shall be examined to determine compliance with requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following table:

TABLE III. Classification of defects.

Categories	Defects *
Major	
101	Rack not free of deformation.
102	All wires not secured at point of intersection.
103	Rack not free of protruding wire ends.
Minor	
201	Rack not stable on flat surface.
202	Rack not free of fins (more than one fin shall be cause for rejection).

* Examination shall not be restricted to the classified defects listed above.

4.3.2 Dimensional examination. The racks shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

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4.4.1 Continuity of coating. The rack shall be thoroughly cleaned, rinsed in running water and immersed in 18% hydrochloric acid (equal parts of concentrated 36% hydrochloric acid and water by weight) at room temperature for half an hour. Examine for bubbling which is indicative of steel dissolving.

NOTE: Any initial bubbling which may be caused by dissolving metallic dust or dirt shall be disregarded. Only bubbling caused by dissolving the improperly coated steel wire shall be taken into account.

4.5 Packaging inspection. The inspection of the unit, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Packaging. Packaging shall be levels A or C, as specified (see 6.2).

5.1.1 Unit of issue. One rack, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package Each rack shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.3 Level C. Each rack shall be packaged in a manner that will afford adequate protection against physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be Level A, B or C, as specified, (see 6.2).

5.2.1 Level A. Twenty-four unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the applicable box specification.

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5.2.2 Level B. Twenty-four unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged racks shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs, shall conform to the applicable carrier's rules and regulations.

5.3 Marking.

5.3.1 Unit packages. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Rack is for general laboratory use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification
(b) National Stock Number (NSN).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 National Stock Number (NSN) coverage. This specification covers the following item as listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6640-00-165-5726	RACK, TEST TUBE, LABORATORY, Epoxy-Coated Wire, 40 Tube

6.4 Limitations. This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification or those which are commercially available, but is intended to cover the types normally procured to meet military requirements.

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6.5 Changes for previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DLA-DM

Project No. 6640-1008

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22 R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-R-37640 RACK, TEST TUBE, LABORATORY, EPOXY-COATED WIRE, 40 TUBE

,ND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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